Access Information

This section provides details about how to connect to the Robot PC from the User PC, including network addresses, credentials, and Web UI access.

Network Information:

- Robot PC IP Address (User PC to Robot PC): `192.168.30.1`
- Robot PC IP Address (Hotspot Connection: `192.168.12.1`
  - Hotspot SSID: `RBY1`
  - Hotspot Password: The password can be viewed after connecting via QR code.

> 🔑 **Connecting via QR Code**
>
> 1. Press the **HOTSPOT button** on the top of the robot's backpack to activate the hotspot.
> 2. A QR code will be displayed; scan it to automatically connect to the network.
>
> - The hotspot will remain active **only while the HOTSPOT button is pressed**.
> - Ensure the button is not pressed again during robot operation, as this will deactivate the hotspot and disrupt communication with the robot.

Web UI Access

- User PC(UPC) to Robot PC(RPC): 192.168.30.1:5173
- Hotspot Connection: 192.168.12.1:5173

Access Credentials

- User PC Username: `nvidia`
- User PC Password: `nvidia`

> ⚠️ **Note**
> The default credentials are standard for Nvidia Jetson devices.

Command Information

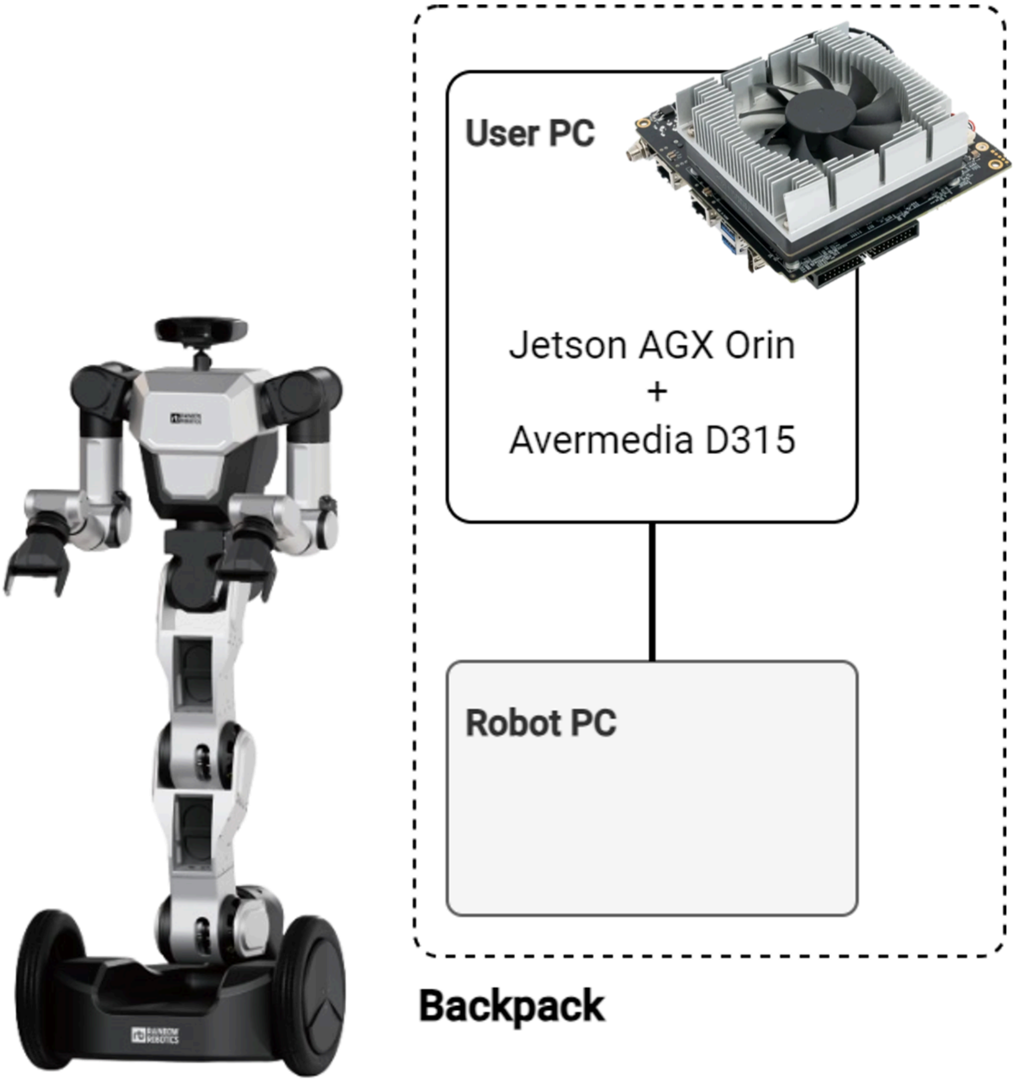
- Command Port: 50051 (Use this port to send commands to the Robot PC.)

Sending Commands to the Robot

From the User PC (UPC)

1. **Direct Command via Code:**
   - Send commands to the Robot PC at `192.168.30.1:50051` from the UPC by writing code that interfaces with the robot.
2. **SSH Connection:**
   - You can also SSH into the UPC and send commands to the Robot PC at `192.168.30.1:50051` .

   > 💡 The UPC only supports **wired connections**, so if you need a wireless solution, you'll need to implement it manually.

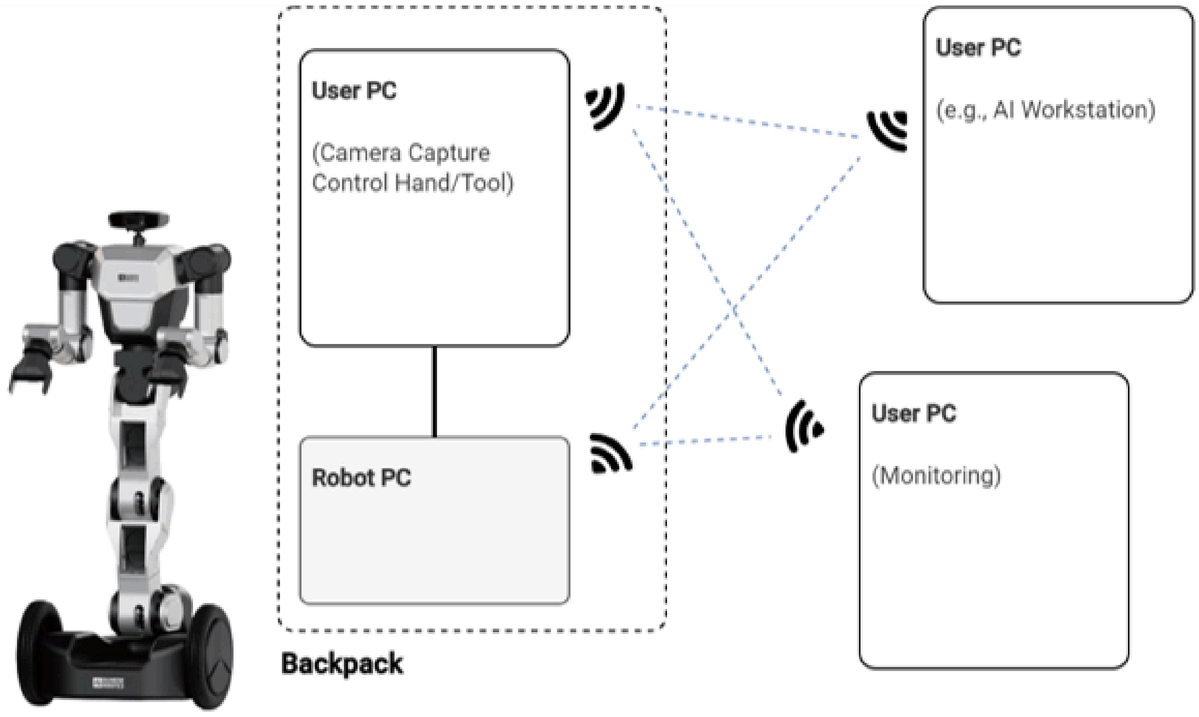


[(Option 1) Using included PC as UPC]

From Another PC(Your own PC)

There are two methods to send commands to the Robot PC from another PC:

1. **Using Hotspot:**
   - Connect the other PC to the same Wi-Fi as the Robot PC via the **HOTSPOT**.
   - Send commands to `192.168.12.1:50051` .
2. **Using a Separate Wi-Fi Network:**
   - Connect the Robot PC and your PC to a different Wi-Fi network.
   - Assign the Robot PC to that Wi-Fi network.
   - Send commands to the **Robot PC's IP address** on the same network, followed by `:50051` for the command port.
   - More detailed instructions on this method will be provided later.



[(Option 2) Using your own PC as UPC]

LiDAR Information

You can access the LiDAR sensors via the following IP addresses:

- **Left LiDAR IP Address:** `192.168.30.10`
- **Right LiDAR IP Address:** `192.168.30.11`

Use these addresses to connect to the respective LiDAR sensors for data collection and monitoring.

Additional Information

- The LiDAR sensors used in this system are **LakiBeam 1L**.
- For more details and documentation, visit the official site: [RichBeam LakiBeam 1L Documentation]

Example Code

A simple example code for printing LiDAR data is available at the following repository: [RainbowRobotics/lakibeam-1l-test]

This example demonstrates basic usage of the LakiBeam 1L LiDAR sensors.

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